

MACHINE LEARNING



Understanding Your Data Before Building Any Model



Table of content

01. Introduction to EDA

What is EDA?

Why is EDA Important?

Dataset Structure

02. Dataset Exploration

Loading Data with Pandas,

Understanding Features, Data

Types, Descriptive Statistics

03. Data Cleaning & Quality Assessment

Missing Values

Duplicate Records

Outliers

04. Data Analysis & Visualization

Univariate Analysis, Bivariate

Analysis, Correlation Analysis,

Basic Visualizations

05. Insights & Next Steps

Interpreting Results

Extracting Business Insights

Preparing Data for Machine Learning

Before You Train a Model...

Your manager hands you 100,000 customer records and asks: “Can you build a model to predict which customers will leave?”

NOT YET.

Before training any model, you need to understand your data. What if half the values are missing, some columns hold incorrect values, records are duplicated, the target is imbalanced, or a feature has the wrong data type?

A model trained on poor-quality data usually produces poor results.

Exploratory Data Analysis (EDA) is the process of examining, summarizing, and visualizing a dataset to understand its characteristics, identify problems, discover patterns, and generate insights — before building any machine learning model.



EDA is like a doctor examining a patient before prescribing medicine.

“Garbage In → Garbage Out (GIGO)”

If you feed poor-quality data into a model, you'll get poor-quality predictions.

Why Do We Perform EDA?



Understand the Dataset

Rows, columns, and the target variable



Detect Missing Values

Can models always handle NaN? Usually no



Detect Duplicates

Duplicate rows distort statistics



Detect Outliers

Extreme values skew averages



Understand Distributions

See the full shape of your data



Discover Relationships

Does area affect price?



Decide on Preprocessing

Scaling, encoding, cleaning



Build Better Models

Clean data → better performance

Spotlight: Missing Values & Duplicates

Detect Missing Values

Age
20
25
NaN
31

Can a machine learning algorithm always handle missing values?

Usually, no — we often need to clean or impute them first.

Detect Duplicate Records

Name	Age
Ahmed	22
Ahmed	22

Duplicates can distort statistics and affect model performance — always check row counts before and after de-duplication.

Spotlight: Outliers & Distributions

Detect Outliers

EMPLOYEE SALARIES

40k

42k

39k

41k

3,000k

Does this last value look normal? Outliers can heavily influence averages and some ML algorithms.

Understand Distributions

AGES

18

19

20

21

22

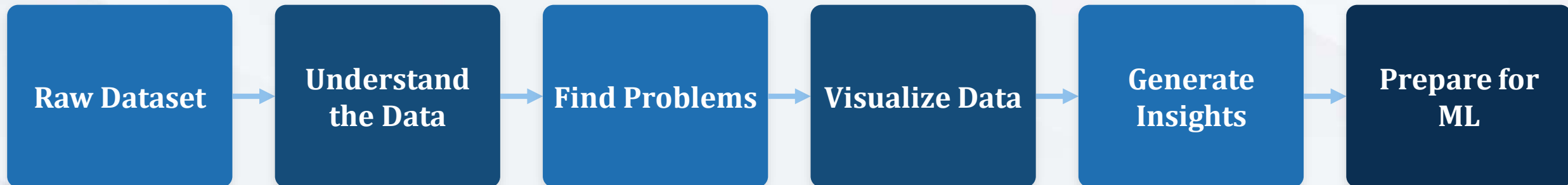
23

24

80

Most people are young, but there's one much older person. Visualizing the distribution reveals this instantly — raw numbers hide it.

The EDA Workflow



Every step feeds the next: you can't visualize data you haven't understood, and you can't preprocess a dataset whose problems you haven't found.

Every Dataset Is Like a Spreadsheet

Area	Bedrooms	Garage	Price
120	3	Yes	250000
150	4	Yes	320000
90	2	No	180000

How many rows? → 3



Rows = Samples

Also called records, instances, or observations. Each row = one real-world object (one house, one student, one customer).



Columns = Variables

Also called features. Each column describes one characteristic of the sample (Area, Bedrooms, Garage, Price).

Features (X) and Target (y)

INDEPENDENT VARIABLES (X)

Features

The input variables the model receives — what we already know.

Area

Bedrooms

Garage

Age

Salary

DEPENDENT VARIABLE (y)

Target

What the model is trying to predict — it depends on the features.

House dataset

Price

Spam dataset

Spam / Not Spam

Medical dataset

Disease / No Disease

Student dataset

Pass / Fail

Numerical Data

Values represented by numbers, where arithmetic operations (mean, median, standard deviation) are meaningful.

Continuous

Can take any value within a range — decimals allowed.

- Height — 175.2 cm
- Weight — 63.75 kg
- Temperature — 36.8°C
- Salary, Distance

Discrete

Can only take specific, countable values — no fractions.

- 2 rooms
- 5 children
- 12 students
- Number of employees, cars

No values like 2.5 children.

Categorical Data

Represents categories or labels instead of quantities.



Nominal

Categories with no natural order.

Gender

Country

Eye Color

Blood Type

“Is Blue greater than Green?” No — there is no ranking.



Ordinal

Categories with a meaningful order, but unequal gaps.

Small

Medium

Large

XL

Education (High School → PhD) and satisfaction (Poor → Excellent) also follow this pattern.

Datetime Data

A single date can be split into several powerful new features.



Purchase Date

2026-07-05

*A single timestamp, waiting
to be unpacked.*

Year

Month

Day

Day of Week

Weekend?

Quarter

Season

*These derived features often improve model performance far more than
the raw date string ever could.*

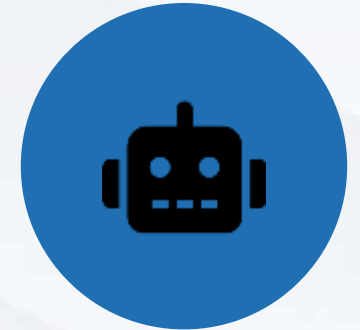
Data Types at a Glance

DATA TYPE	EXAMPLES	MEAN?	VISUALIZATION
Continuous Numerical	Height, Salary	✓ Yes	Histogram, Box Plot
Discrete Numerical	Number of Rooms	✓ Usually	Bar Chart, Histogram
Nominal Categorical	Gender, Country	✗ No	Count Plot, Bar Chart
Ordinal Categorical	Small, Medium, Large	✗ No	Ordered Bar Chart
Datetime	Purchase Date	✗ Directly	Line Chart, Time Series

Different data types need different preprocessing and different charts

Identify the Data Type

VARIABLE	TYPE
Age	Continuous Numerical
Number of Students	Discrete Numerical
Gender	Nominal Categorical
T-shirt Size	Ordinal Categorical
Purchase Date	Datetime
House Price	Continuous Numerical



Once we receive a dataset, what is the first thing we should do?

Not train a model. Not plot graphs. First, we inspect the dataset.

Step 1-2: Import and Load

Pandas is the most popular Python library for working with tabular data.

```
import pandas as pd

df = pd.read_csv("data.csv")
```

- pandas loads, cleans, analyzes, and manipulates data efficiently
- `read_csv()` reads a Comma-Separated Values file
- The result is stored in a DataFrame



What Is a DataFrame?

A 2-dimensional table of rows and columns — similar to an Excel spreadsheet.

Age	Salary	Gender
25	50000	Female
31	62000	Male
42	58000	Male

Step 3-4: First and Last Rows

Always look at the edges of your data before anything else.

`df.head()`

Returns the first 5 rows.

- Quickly inspect the dataset
- Verify it loaded correctly
- See column names
- Get a first impression of the data

`df.tail()`

Returns the last 5 rows.

- Whether the file ended correctly
- Final records in the dataset
- Unexpected values at the end

Step 5-6: Shape and Column Names

`df.shape`

```
(1000, 8)
```

- 1000 rows (samples)
- 8 columns (features)

`df.columns`

```
Index(['Age', 'Gender',  
      'Salary', 'Purchased'])
```

Useful for:

- Understanding available features
- Detecting misspelled names
- Selecting columns later

Step 7: Dataset Information

One of the most useful EDA functions — a full structural summary.

```
df.info()
```

```
RangeIndex: 1000 entries  
Data columns: 5
```

```
Age           int64  
Salary        float64  
Gender        object  
Purchased     int64
```

What info() Tells You

Number of Rows

How many samples exist in total.

Data Types

int64, float64, object, bool, datetime.

Missing Values

1000 rows but 995 non-null → 5 missing.

Memory Usage

Larger datasets require more RAM.

Understanding Data Types

Each column has a type that determines how pandas treats it.

int64

Whole numbers, e.g.
Age

float64

Decimal numbers,
e.g. Salary

object

Text or categorical
values

bool

True / False values



Why It Matters

Data types determine which operations and statistics apply — you can average a float column, but not a text column.

Step 8: Statistical Summary

```
df.describe()
```

Statistic	Meaning
count	Number of non-missing values
mean	Average
std	Standard deviation (spread of the data)
min	Smallest value
25%	First quartile
50%	Median
75%	Third quartile
max	Largest value

Example: Age

```
Mean = 35
```

```
Min = 18
```

```
Max = 70
```

Interpretation:

Most people in this dataset are around 35 years old.

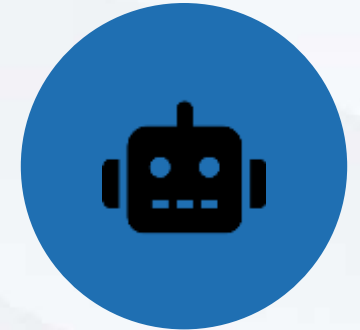
Step 9: Random Samples & Recap



`df.sample(5)`

Returns five random rows — the first rows aren't always representative, so random sampling gives a more balanced view.

Function	Purpose
<code>head()</code>	First rows
<code>tail()</code>	Last rows
<code>shape</code>	Rows and columns count
<code>columns</code>	Column names
<code>info()</code>	Dataset overview
<code>describe()</code>	Statistical summary
<code>sample()</code>	Random observations



Understanding Features

understand each feature individually

Unique Values

```
df["Gender"].unique()
```

```
['Male', 'Female']
```

Returns every distinct value in the column.

Useful for understanding categories and detecting spelling inconsistencies:

```
Male, male, M,  
Female, female
```

5 unique values — but probably only 2 real categories.



```
df["Gender"].nunique()
```

```
2
```

Returns the number of unique values, while unique() returns the values themselves.

Value Counts and Frequency

How many times does each category appear?

```
df["Gender"].value_counts()
```

Male	650
Female	350

Shows raw counts — useful for checking class balance and finding dominant categories.

```
df["Gender"].value_counts(  
    normalize=True)
```

Male	0.65
Female	0.35

Returns proportions instead of counts — multiply by 100 to read as percentages: Male 65%, Female 35%.

Why Is This Useful?

Suppose your target variable is Purchased:

Yes
Yes
Yes
Yes
No

$$95\% = \text{Yes} \cdot 5\% = \text{No}$$

Your dataset is imbalanced. This affects model performance — a model can reach 95% accuracy just by always predicting “Yes.”

Yes — 95%

No — 5%

Why Do Missing Values Occur?



Human Error

Someone forgot to enter information.



Skipped Fields

Optional fields like phone number are left blank.



Sensor Failure

A temperature sensor stopped working → NaN.



Data Transfer Issues

Some records were lost during collection.

Detecting Missing Values

```
df.isnull().sum()
```

Age	3
Salary	10
Gender	0

- isnull() checks every cell
- sum() counts the missing values per column



Ask Yourself

Suppose Salary has 400 missing values out of 500 rows.

Would you keep this feature?

Probably not.

Common Solutions



Remove Rows

When only a few rows are affected (e.g. 3 out of 10,000). Safe to remove.



Remove Columns

When a column has too many missing values (e.g. 90%). Probably useless.



Fill with Mean

Best for numerical data without extreme outliers.



Fill with Median

Better when outliers exist, e.g. one salary of 500k among 30k values.



Fill with Mode

Best for categorical features — fill with the most frequent category.

Detecting and Removing Duplicates

Duplicate rows can bias statistics, skew averages, and give some samples more influence during training.



Detect Duplicates

```
df.duplicated().sum()
```

25

Meaning: there are 25 duplicated rows in the dataset.



Remove Duplicates

```
df = df.drop_duplicates()
```

```
# or  
df.drop_duplicates(inplace=True)
```

`inplace=True` updates the existing DataFrame directly instead of returning a new one.

Outliers: The First Question

“Suppose we're analyzing employee salaries. Does Employee E look normal?”

Employee	Salary (\$)
A	45,000
B	48,000
C	50,000
D	47,000
E	5,000,000

Definition

An outlier is a value that is significantly different from the rest of the data — extremely high or extremely low.

Example — ages:

18, 19, 20, 21, 22, 150

150 years old is probably an error.

Employee E's salary is an outlier — a value that stands far apart from everyone else.

Why Outliers Matter: They Distort the Mean

Without Outlier

40 42 41 39 43

Average = 41

The average represents the typical value well.

With Outlier

40 42 41 39 500

Average $\approx$ 132.4

The average is much larger — it no longer represents the typical employee.

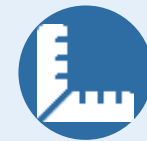
Why Outliers Matter: Models & Scaling



They Affect ML Models

Some algorithms are sensitive to outliers:

- Linear Regression — sensitive
- K-Means Clustering — sensitive
- KNN (distance-based) — sensitive
- Decision Trees — generally more robust



They Affect Scaling

If one value is extremely large, Min-Max Scaling compresses all remaining values into a tiny range — losing meaningful spread among the majority of the data.

Are All Outliers Bad? & How to Detect Them

Don't remove outliers automatically — first, understand why they exist.

A \$5B salary among billionaires may be valid; the same value among university students is probably an error.



Box Plot (Most Common)

Minimum, Q1, Median, Q3, Maximum. Points outside the whiskers are potential outliers.



IQR Method

Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are considered potential outliers.



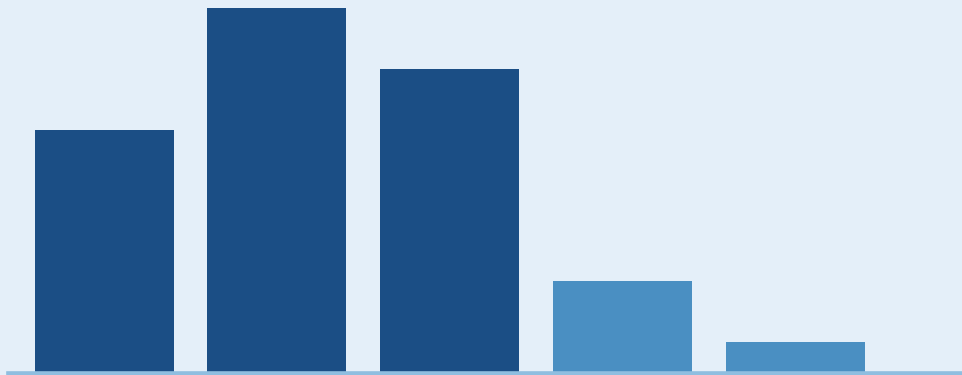
Z-Score

Useful when data is approximately normally distributed.

Univariate Analysis: Numerical Features

“Uni” means one — we analyze one feature at a time.

Histogram — Age Distribution



Mean vs. Median

- Mean — average value; sensitive to outliers.
- Median — middle value; much more robust.

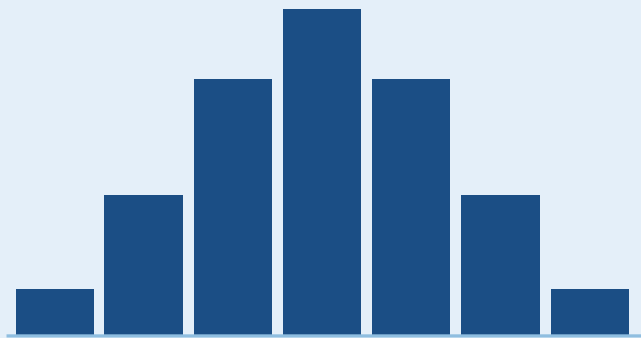
Example: 20, 21, 22, 23, 500 — the mean is distorted, but the median still represents the center well.

Where are most observations? Symmetric? Multiple peaks? Any outliers?

Spread: if everyone earns around \$50,000, spread is small. If salaries range from \$20,000 to \$500,000, spread is large. Measured with **range** and **standard deviation**.

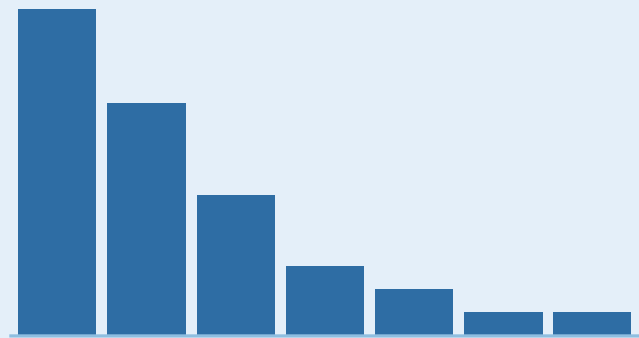
Skewness: The Shape of a Distribution

Symmetric



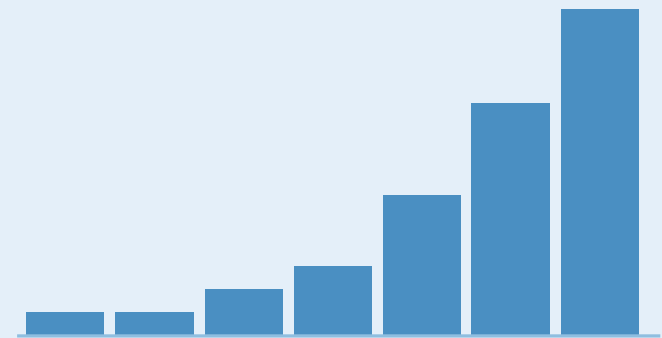
Mean $\approx$ Median. Bell-shaped and balanced on both sides.

Right Skew



Long tail on the right — caused by a few large values (e.g. salary, house prices).

Left Skew



Long tail on the left. Less common than right skew.

Many statistical techniques assume approximately normal (symmetric) data — such as heights or measurement errors.

Univariate Analysis: Categorical Features



Count Plot

Shows how many observations belong to each category — easy to compare frequencies.

Male



Female



Which category is most common? Which is least common?



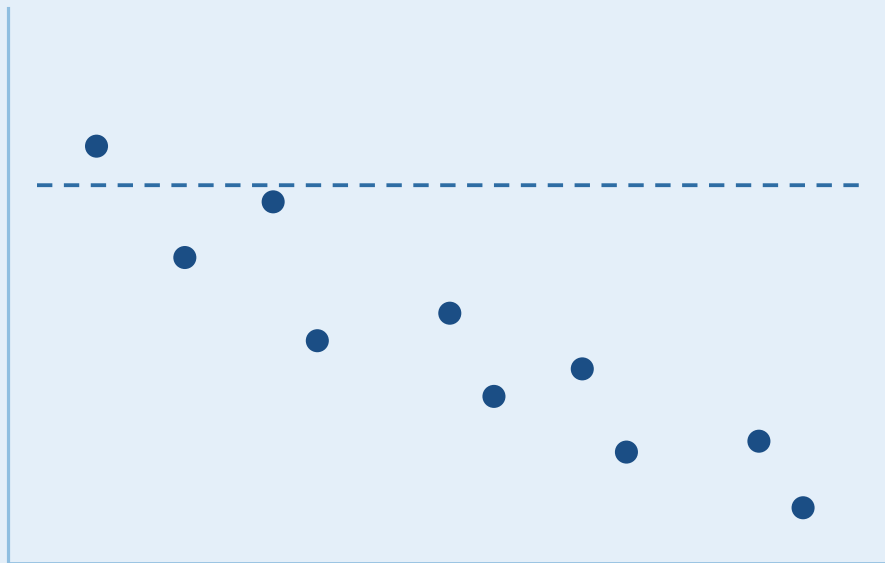
Bar Chart

Similar to a count plot, but typically used when the data is already aggregated (e.g. totals per category from a summary table, rather than raw rows to count).

Bivariate Analysis: Numerical vs. Numerical

“Does studying more hours improve exam scores?”

Scatter Plot — Hours Studied vs. Exam Score



Correlation

Positive

Hours $\uparrow$ $\rightarrow$ Score $\uparrow$

Negative

Price $\uparrow$ $\rightarrow$ Demand $\downarrow$

No Correlation

No clear pattern between variables

Bivariate Analysis: Categorical Relationships

Categorical vs. Numerical

Example: Gender vs. Salary — do males and females have similar salary distributions?

- Box Plot — compares medians, spread, and detects outliers. Very useful.
- Violin Plot (optional) — like a box plot, but also shows the data density.

Categorical vs. Categorical

Example: Gender vs. Purchased

```
pd.crosstab(df["Gender"],  
df["Purchased"])
```

	No	Yes
Male	35	62
Female	28	70

A heatmap makes the crosstab easier to read — darker cells indicate higher counts.

Correlation Matrix

`df.corr(numeric_only=True)` — compute all pairwise correlations at once

	Age	Salary	Experience
Age	1.00	0.82	0.91
Salary	0.82	1.00	0.78
Experience	0.91	0.78	1.00

Reading the Heatmap

- Dark → strong relationship
- Light → weak relationship
- +1 → perfect positive (both increase together)
- -1 → perfect negative (one up, one down)
- 0 → no linear relationship



Correlation ≠ Causation

Ice cream sales increase. Drowning incidents increase. Does ice cream cause drowning? No — both rise in summer because of a third factor: hot weather.

Basic Visualizations

Chart	Best For	Main Purpose	Example Question
Histogram	Numerical	Distribution	What is the age distribution?
Box Plot	Numerical	Outliers & spread	Are there unusual salaries?
Bar Chart	Categories	Compare counts/values	Which category is most common?
Pie Chart	Categories	Proportions (few categories)	What share does each group hold?
Scatter Plot	Two numerical vars	Relationship	Does experience relate to salary?
Line Plot	Time series	Trends over time	How did monthly sales change?
Heatmap	Matrix data	Correlation or crosstab	Which variables are strongly correlated?

Insights: The Real Goal of EDA

“EDA is not about creating beautiful graphs — it's about answering questions.”

WEAK

“The average salary is \$50,000.” — This is just reporting a number.

STRONG

“Most employees earn between \$40,000 and \$60,000, but a few extremely high salaries create a right-skewed distribution, making the mean higher than the median.”

STRONG

“Most customers are between 25 and 35 years old, suggesting the company's primary audience is young adults.”

Thank you